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MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : M-401

MATHEMATICS - III

Time Allotted : 3 *Hours* *Full Marks* : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

GROUP – A

(Multiple Choice Type Questions)

1. Choose the correct alternatives for any *ten* of the following : $10 \times 1 = 10$

i) If A and B are two events with $P(A) = 0{\cdot}4$, $P(B) = 0{\cdot}3$ and $P(A \cap B) = 0{\cdot}2$, then $P(A \cap B^C)$ is

a) 0·1 b) 0·2

c) 0·3 d) 0·4.

ii) A fair die is thrown. The probability that either an odd number or a number greater than 4 will turn up is

a) $\frac{2}{5}$ b) $\frac{3}{7}$

c) $\frac{2}{7}$ d) $\frac{2}{3}$.

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iii) The variance of a rectangular uniform distribution with parameters a and b is

a) $\frac{b-a}{2}$ b) $\frac{(b-a)^2}{12}$

c) $\frac{(a+b)^2}{12}$ d) $\frac{(b-a)^2}{2}$.

iv) A random variable X has the following probability density function :

$$f(x) = \begin{cases} kx, & 1 \le x \le 2 \\ 0, & \text{otherwise} \end{cases}$$

The value of k is

a) 1 b) 2

c) 2/3 d) none of these.

v) If a is an element of a group G and order of a is 35, then the order of a^{10} is

a) 5 b) 7

c) 35 d) none of these.

vi) Inverse of the permutation $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}$ is

a) $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix}$ b) $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}$

c) $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}$ d) none of these.

vii) The number of generators of an infinite cyclic group is

a) 1 b) 2

c) 0 d) infinite.

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viii) The mean and standard deviation of standard normal variable are

a) 0, − 1 b) 1, 1

c) 0, 1 d) 1, 0.

ix) The statistic t is said to be an unbiased estimator of a population parameter θ when

a) $E(t) = \theta$ b) $E(t^2) = \theta$

c) $E(t^2) = [E(\theta)]^2$ d) $[E(t)]^2 = [E(\theta)]^2$.

x) A complete graph is called Kuratowski's first graph if it has

a) 5 vertices b) 6 vertices

c) 4 vertices d) 8 vertices.

xi) The number of generators of a cyclic group of order 7 is

a) 2 b) 6

c) 7 d) 5.

xii) An element a in a ring R is zero divisor if

a) $a.b = 0$

b) $a.b \neq 0$ for all element b in R

c) $a.b = 0$ for some non-zero element b in R

d) none of these.

xiii) If C_{97} be a circuit with 97 number of vertices, then chromatic number of C_{97} is

a) 97 b) 2

c) 3 d) none of these.

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GROUP – B

(Short Answer Type Questions)

Answer any *three* of the following. 3 × 5 = 15

2. Determine the mean and variance of exponential distribution.

3. Let G be a simple connected planar graph with e edges and f regions. Then prove that $e \geq \frac{3}{2}f$.

4. Show by Tchebycheff's inequality that in 2000 throws with a coin, the probability that the number of heads lies between 900 and 1100 is at least $\frac{19}{20}$.

5. Prove that a group G is Abelian, iff $(ab)^2 = a^2b^2$ for all $a, b \in G$.

6. Show that the permutation $\begin{pmatrix} 1 & 23 & 4 & 5 \\ 3 & 21 & 5 & 4 \end{pmatrix}$ is even.

7. An urn contains 10 white and 3 black balls, while another urn contains 3 white and 5 black balls. Two balls are drawn from the first urn and put into the second urn and then a ball is drawn from the latter. What is the probability that it is a white ball ?

GROUP – C

(Long Answer Type Questions)

Answer any *three* of the following. 3 × 15 = 45

8. a) Show that the group $(Z_5, +)$, *i.e.* the additive group of all integers modulo 5 is cyclic. Find all generators of Z_5.

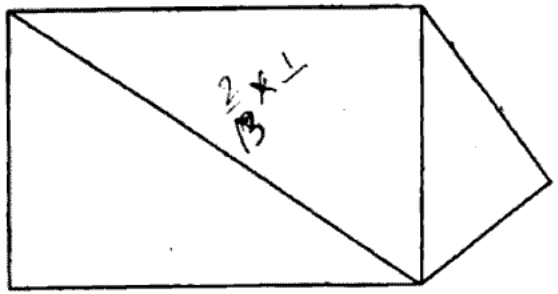
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b) Draw the dual of the graph.



c) Prove that order of each subgroup of a finite group is divisor of the order of the group. 5 + 5 + 5

9. a) Show by Tchebyshev's inequality that if a die is thrown 3,600 times, the probability of number of sixes that lies between 550 and 650 is at least.

b) The mean of a normal distribution is 50 and 5% of the values are greater than 60. Find the standard deviation of the distribution (Area under standard normal curve between $z = 0$ and $z = 1{\cdot}64$ is 0·45)

c) If a population has Poisson distribution with parameter λ, then show that the sample mean is the maximum likelihood estimate of λ. 5 + 5 + 5

10. a) Define planar graph. Construct a planar graph with 6 vertices.

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b) If G be a connected graph with n vertices, e edges and r faces, prove that $n - e + r = 2$.

c) If $x_1, x_2, x_3, x_4, x_5, x_6$ be an independent simple random sample from a normal population with unknown variance σ^2, find K so that $K\left[(X_1 - X_2)^2 + (X_3 - X_4)^2 + (X_5 - X_6)^2\right]$ is an unbiased estimator of σ^2. 5 + 5 + 5

11. a) The weights in pound of a sample of 12 packets of butter are 13, 7, 22, 15, 12, 18, 14, 21, 8, 17, 10, 23 taken at random from its population having standard deviation 5. Find 95% confidence interval for the mean of the population. [Given $Z_{0.05} = 1{\cdot}96$]

b) If H and K are two subgroups of a group G, then prove that HK is also subgroup of G if and only if $HK = KH$.

c) Let $f: G \to G'$ be a homomorphism. Show that f is one-to-one if and only if Ker $f = \{e\}$. 5 + 5 + 5

12. a) Prove that $H = \left\{ \begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix} \mid x \in \mathbb{R},\ x \neq 0 \right\}$ forms a normal subgroup of $GL(2, \mathbb{R})$, the group of all real non-singular 2×2 matrices.

b) Prove that the intersection of two subrings is a subring.

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c) A random sample with observations 65, 71, 64, 71, 70, 69, 64, 63, 67, 68 is drawn from a normal population with standard deviation. Test the hypothesis that population mean is 69 at 1% level of significance.

[Given that $P(0 < z < 2{\cdot}58) = 0{\cdot}495$] 5 + 5 + 5

13. a) Prove that every group of prime order is commutative.

b) Show that for the exponential distribution,

$$f(x) = \frac{1}{\sigma} e^{\left(-\frac{x}{\sigma}\right)}, \quad 0 < x < \infty$$

mean and standard deviation both equal to σ.

c) The following figures show the distribution of digits in numbers chosen at random from a telephone directory :

Digits	0	1	2	3	4	5	6	7	8	9	Total
Frequency	1026	1107	997	966	1075	933	1107	972	964	853	10,000

Test whether the digits may be taken to occur equally frequently in the directory.

[The tabulated $\chi^2_{0.05}$ for 9 d.f = 16·919] 5 + 5 + 5

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